

IA 642 Defensive Security

Midterm Project: Memory Forensics

WannaCry and Stuxnet from RAM

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Lab overview

This midterm investigates two Windows XP classroom memory images: WannaCry (`wcry.vmem`, May 2017) and Stuxnet (`stuxnet.vmem`, June 2011). Analysis uses Volatility 2.6 plugins equivalent to the course SIFT `vol.py` workflow. Values below come from live plugin runs on local copies extracted from `memory_dump.zip`. The Symantec *W32.Stuxnet Dossier* v1.3 is required reading for Parts B and C. MITRE ATT&CK software pages S0366 (WannaCry) and S0603 (Stuxnet) are used for technique mapping.

Environment and tooling

- **Evidence:** `wcry.vmem` and `stuxnet.vmem` from Canvas Week 05 Midterm / `memory_dump.zip`; dossier PDF from the same module.
- **SIFT Workstation:** Course `SIFT-Workstation.ova` imported into VirtualBox as VM `SIFT-Workstation` (Ubuntu 20.04, 4 GB RAM, 4 vCPU, NAT NIC host port 2225→guest 22, snapshot `clean-after-import`). Long-mode/64-bit enabled after OVA import. Volatility 2.6.1 is available inside SIFT as `/usr/local/bin/vol.py` (Module 02 workflow).
- **Analysis runs:** Graded plugin excerpts use Volatility 2.6 profile `WinXPSP2x86`. The same dumps were re-run inside SIFT with `vol.py` (`pslist/pstree/psscan/cmdline/dlllist/files-can/modules/modscan/ssdt/driverscan/connections/sockets`) and matched the host findings (WannaCry `tasksche.exe`/PID 1940 and `@WanaDecryptor@`/PID 740; Stuxnet triple `lsass.exe` PIDs). Volatility 3 was not used for graded XP net answers.
- **Screenshots:** Question-labeled terminal figures (A1, A2, A3, A5, B2, B5/B6, B7, B8, B10) plus SIFT Workstation GUI captures (`fig_SIFT_*`).
- **Safety:** Lab IOCs defanged in this report; VirusTotal used hash-only (no sample upload); SIFT kept on NAT / isolated for analysis.
- **Public evidence repo:** Analysis outputs, figures, and this PDF (memory dumps excluded; SHA-256 hashes only): <https://github.com/unaisshazan/ia642-week05-midterm-memory-forensics>

Evidence Intake and Scope

All timestamps in analysis are UTC unless noted as host-local from `imageinfo`.

Artifact	SHA-256	Size (bytes)
wcry.vmem	76E8BE1A3761878325FDFF39A5AB1FF84922A0B18947E5268DD9175795AD2BF0	536870912
stuxnet.vmem	5F19FF1333FC3901FBF3FAFB50D2ADB0C495CF6D33789E5A959499A92AEEFE77	536870912
memory_dump.zip	0F73E7C15F16A6073EDE0F228D33F53996953FAC63844E33AF5253478994601C	n/a
dossier PDF	85F536DB77070DC2228824D27DCBDD07DB73C36E2652DCCB9F99569C3D1AEA3E	n/a

Chain of custody: Analyst Unais Ali; received via Canvas Week 05 Midterm module; intake hashes match final analysis copies (no transformation before Volatility 2.6 runs).

Part A: WannaCry

A1. Identify the image and choose a profile

Hypothesis. WinXPSP2x86 is the correct operational profile: `imageinfo` suggests WinXPSP2x86 and WinXPSP3x86; I instantiate WinXPSP2x86 when Image Type reports SP3. DTB/KDBG must align before process parsing.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" imageinfo
```

Supporting output

```
Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
AS Layer1 : IA32PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\temp\midterm\dumps\wcry.vmem)
PAE type : No PAE
DTB : 0x39000L
KDBG : 0x8054cf60L
Number of Processors : 1
Image Type (Service Pack) : 3
KPCR for CPU 0 : 0xffdff000L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2017-05-12 21:26:32 UTC+0000
Image local date and time : 2017-05-13 02:56:32 +0530
```

```
A1 | vol26.exe imageinfo | wcry.vmem

Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
AS Layer1 : IA32PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\temp\midterm\dumps\wcry.vmem)
PAE type : No PAE
DTB : 0x39000L
KDBG : 0x8054cf60L
Number of Processors : 1
Image Type (Service Pack) : 3
KPCR for CPU 0 : 0xffdff000L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2017-05-12 21:26:32 UTC+0000
Image local date and time : 2017-05-13 02:56:32 +0530
```

Figure 1: A1 screenshot: `imageinfo` on `wcry.vmem` (live Volatility 2.6).

Interpretation. Suggested profiles include WinXPSP2x86 and WinXPSP3x86 (instantiated

WinXPSP2x86). Address space is IA32PagedMemory with PAE No and Image Type (Service Pack) 3. Image date and time 2017-05-12 21:26:32 UTC marks capture near the end of encryption activity. Local time +0530 reflects host timezone configuration in the VM, not attacker geography. Verification: `pslist` succeeds under both SP2 and SP3 profile suggestions; WinXPSP2x86 retained for consistency.

A2. The malicious lineage

Hypothesis. WannaCry detonation chain runs under `explorer.exe` (interactive session), not as an SMB-spawned System child visible in this snapshot.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
pstree
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
cmdline
```

Supporting output

Name	Pid	PPid	Thds	Hnds	Time
-----	-----	-----	-----	-----	-----
0x823c8830:System UTC+0000	4	0	51	244	1970-01-01 00:00:00
. 0x82169020:smss.exe UTC+0000	348	4	3	19	2017-05-12 21:21:55
.. 0x8216e020:winlogon.exe UTC+0000	620	348	23	536	2017-05-12 21:22:01
... 0x82191658:lsass.exe UTC+0000	676	620	23	353	2017-05-12 21:22:01
... 0x821937f0:services.exe UTC+0000	664	620	15	265	2017-05-12 21:22:01
.... 0x821af7e8:svchost.exe UTC+0000	1024	664	79	1366	2017-05-12 21:22:03
..... 0x81f747c0:wuaucflt.exe UTC+0000	1768	1024	7	132	2017-05-12 21:22:52
..... 0x81fea8a0:wscntfy.exe UTC+0000	1168	1024	1	37	2017-05-12 21:22:56
.... 0x821bea78:svchost.exe UTC+0000	1152	664	10	173	2017-05-12 21:22:06
.... 0x81fb95d8:svchost.exe UTC+0000	260	664	5	105	2017-05-12 21:22:18
.... 0x821b5230:svchost.exe UTC+0000	904	664	9	227	2017-05-12 21:22:03
... (22 lines total; truncated)					
--- cmdline (filtered) ---					
Command line : \SystemRoot\System32\smss.exe					
Command line : C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv: UserServerDllInitialization,3 ServerDll=winsrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads=16					
Command line : winlogon.exe					
Command line : C:\WINDOWS\system32\services.exe					
Command line : C:\WINDOWS\system32\lsass.exe					
Command line : C:\WINDOWS\system32\svchost -k DcomLaunch					
Command line : C:\WINDOWS\system32\svchost -k rpcss					

```

Command line : C:\WINDOWS\System32\svchost.exe -k netsvcs
Command line : C:\WINDOWS\system32\svchost.exe -k NetworkService
Command line : C:\

```

A2 | vol26.exe pstree | wcry.vmem

Name	Pid	PPid	Thds	Hnds	Time
0x823c8830:System	4	0	51	244	1970-01-01 00:00:00 UTC+0000
. 0x82169020:smss.exe	348	4	3	19	2017-05-12 21:21:55 UTC+0000
.. 0x8216e020:winlogon.exe	620	348	23	536	2017-05-12 21:22:01 UTC+0000
... 0x82191658:lsass.exe	676	620	23	353	2017-05-12 21:22:01 UTC+0000
... 0x821937f0:services.exe	664	620	15	265	2017-05-12 21:22:01 UTC+0000
.... 0x821af7e8:svchost.exe	1024	664	79	1366	2017-05-12 21:22:03 UTC+0000
.... 0x81f747c0:wuauc1t.exe	1768	1024	7	132	2017-05-12 21:22:52 UTC+0000
.... 0x81fea8a0:wscntfy.exe	1168	1024	1	37	2017-05-12 21:22:56 UTC+0000
.... 0x821bea78:svchost.exe	1152	664	10	173	2017-05-12 21:22:06 UTC+0000
.... 0x81fb95d8:svchost.exe	260	664	5	105	2017-05-12 21:22:18 UTC+0000
.... 0x821b5230:svchost.exe	904	664	9	227	2017-05-12 21:22:03 UTC+0000
.... 0x8203b7a8:svchost.exe	1084	664	6	72	2017-05-12 21:22:03 UTC+0000
.... 0x821e2da0:spoolsv.exe	1484	664	14	124	2017-05-12 21:22:09 UTC+0000
.... 0x82010020:alg.exe	544	664	6	101	2017-05-12 21:22:55 UTC+0000
... (truncated for figure)					

Figure 2: A2 screenshot: pstree lineage on wcry.vmem.

Interpretation. explorer.exe PID 1636 lists PPID 1608 (parent not in pslist: exited parent, common after userinit). tasksche.exe PID 1940 parent 1636 at 21:22:14 UTC; @WanaDecryptor@ PID 740 parent 1940 at 21:22:22 UTC. EPROCESS ImageFileName is 16-character truncated; full path recovered via cmdline as C:/Intel/ivecuqmanpnirkt615/tasksche.exe (and @WanaDecryptor@.exe). Root under explorer supports user-session detonation rather than worm-only remote spawn as parent.

A3. pslist versus psscan

Hypothesis. Disagreement between pslist and psscan for WannaCry helpers reflects exited processes removed from ActiveProcessLinks, not DKOM rootkit hiding.

Exact command

```

"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
psxview
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
psscan

```

Supporting output

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd
	ExitTime								
0x02218da0	tasksche.exe	1940	True	True	False	True	True	True	True
0x021d9da0	explorer.exe	1636	True	True	False	True	True	True	True
0x01fde308	@WanaDecryptor@	740	True	True	False	True	True	True	True

A3 | vol26.exe psxview | wcry.vmem

Offset(P)	Name	PID	pslist	psscan	thrdproc	pspcid	csrss	session	deskthrd	ExitTime
0x0216e020	winlogon.exe	620	True	True	False	True	True	True	True	
0x0221a2c0	svchost.exe	836	True	True	False	True	True	True	True	
0x02218da0	tasksche.exe	1940	True	True	False	True	True	True	True	
0x01fb95d8	svchost.exe	260	True	True	False	True	True	True	True	
0x01f747c0	wuauc1t.exe	1768	True	True	False	True	True	True	True	
0x021d9da0	explorer.exe	1636	True	True	False	True	True	True	True	
0x021af7e8	svchost.exe	1024	True	True	False	True	True	True	True	
0x021bea78	svchost.exe	1152	True	True	False	True	True	True	True	
0x02231da0	ctfmon.exe	1956	True	True	False	True	True	True	True	
0x01fea8a0	wscntfy.exe	1168	True	True	False	True	True	True	True	
0x01fde308	@WanaDecryptor@	740	True	True	False	True	True	True	True	
0x021937f0	services.exe	664	True	True	False	True	True	True	True	
... (truncated for figure)										

Figure 3: A3 screenshot: psxview showing exited helpers (pslist False / psscan True).

Interpretation. psxview shows `taskse.exe` (536) and `taskdl.exe` (860) False in pslist, True in psscan, with ExitTime 2017-05-12 21:26:23 UTC. Exited `@WanaDecryptor@` instances also appear only in psscan with ExitTime. Kernel mechanism: pslist walks the linked EPROCESS list; psscan searches pool tags for `_EPROCESS` remnants. ExitTime populated means terminated, not stealth-hidden. On a clean system the same disagreement can occur for recently exited processes.

A4. UTC timeline and ATT&CK S0366 mapping

Hypothesis. Encryptor and helper processes cluster just before image acquisition; map to S0366 impact and recovery-inhibition techniques.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
pstree
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
psxview
```

Supporting output

```
2017-05-12 21:22:14 UTC tasksche.exe (1940)
2017-05-12 21:22:22 UTC @WanaDecryptor@ (740)
2017-05-12 21:25:53 UTC @WanaDecryptor@ (424) ExitTime (psscan)
2017-05-12 21:26:23 UTC taskse/taskdl exited; @WanaDecryptor@ (576) ExitTime
2017-05-12 21:26:32 UTC image acquisition (imageinfo)
```

Interpretation. Memory-observed mapping (MITRE ATT&CK S0366): T1486 Data Encrypted for Impact (`@WanaDecryptor@`, `tasksche` orchestration); T1490 Inhibit System Recovery (shadow deletion helpers `taskse/taskdl`). Timeline shows encryption activity within about four minutes of `tasksche` start, immediately prior to dump.

A5. Network artifacts and MS17-010 context

Hypothesis. TCP 445 listener on System PID 4 is consistent with SMB exposure used by WannaCry propagation (CVE-2017-0144 / MS17-010); absence of connscan rows for WannaCry PIDs does not disprove spread attempts.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86 sockets
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86 connscan
```

Supporting output

Offset(V)	PID	Port	Proto	Protocol	Address	Create Time
0x8223d3b8	4	138	17	UDP	192.168.56.101	2017-05-12 21:21:56 UTC+0000
0x81fb4890	4	0	47	GRE	0.0.0.0	2017-05-12 21:22:55 UTC+0000
0x81f707c0	1024	1025	17	UDP	127.0.0.1	2017-05-12 21:22:53 UTC+0000
0x81f74c18	676	500	17	UDP	0.0.0.0	2017-05-12 21:22:53 UTC+0000
0x82188658	1024	123	17	UDP	192.168.56.101	2017-05-12 21:22:53 UTC+0000
0x8219b3b8	4	445	6	TCP	0.0.0.0	2017-05-12 21:21:55 UTC+0000
0x821824b0	904	135	6	TCP	0.0.0.0	2017-05-12 21:22:03 UTC+0000
0x81ffb890	4	1027	6	TCP	0.0.0.0	2017-05-12 21:22:55 UTC+0000
0x81f6a360	1024	123	17	UDP	127.0.0.1	2017-05-12 21:22:53 UTC+0000
... (19 lines total; truncated)						
--- connscan ---						
Offset(P)	Local Address		Remote Address		Pid	
-----	-----		-----		---	

Offset(V)	PID	Port	Proto	Protocol	Address	Create Time
0x8223d3b8	4	138	17	UDP	192.168.56.101	2017-05-12 21:21:56 UTC+0000
0x81fb4890	4	0	47	GRE	0.0.0.0	2017-05-12 21:22:55 UTC+0000
0x81f707c0	1024	1025	17	UDP	127.0.0.1	2017-05-12 21:22:53 UTC+0000
0x81f74c18	676	500	17	UDP	0.0.0.0	2017-05-12 21:22:53 UTC+0000
0x82188658	1024	123	17	UDP	192.168.56.101	2017-05-12 21:22:53 UTC+0000
0x8219b3b8	4	445	6	TCP	0.0.0.0	2017-05-12 21:21:55 UTC+0000
0x821824b0	904	135	6	TCP	0.0.0.0	2017-05-12 21:22:03 UTC+0000
0x81ffb890	4	1027	6	TCP	0.0.0.0	2017-05-12 21:22:55 UTC+0000
0x81f6a360	1024	123	17	UDP	127.0.0.1	2017-05-12 21:22:53 UTC+0000
0x81f7fe98	676	0	255	Reserved	0.0.0.0	2017-05-12 21:22:53 UTC+0000
... (truncated for figure)						

Figure 4: A5 screenshot: sockets showing TCP/445 on System PID 4.

Interpretation. sockets shows TCP port 445 bound to PID 4 (System) and host IP 192.168[.].56[.].101 (VirtualBox host-only). No WannaCry PID owns connscan-visible connections at capture time. MS17-010 (EternalBlue) is the documented propagation vector for S0366; empty connection tables may reflect timing, closed sockets, or lab isolation, not evidence that the malware family lacks network capability.

A6. Safety, defanged IOCs, and kill-switch reasoning

Hypothesis. Kill-switch DNS logic (tasksche/mssecsvc) can explain global slowdown when the domain registered May 12 2017; this VM was still encrypted (unreachable DNS, race after encryptor start, or local detonation under explorer).

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
strings / yarascan (supporting)
Manual IOC review with defanging for report.
```

Supporting output

```
Defanged IOCs (representative):
URL: hxxp://www[.]iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea[.]com
BTC: 12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw ; 13AM4VW2dhxYgXeQepoHkHSQuy6NgaEb94
Kill-switch: malware checks domain; if resolves, exit. Registration slowed outbreak globally.
```

Interpretation. Host encrypted anyway, so the kill-switch likely failed from this VM (isolated lab DNS), the check raced after encryption began, or the sample path ignored the switch. Lineage under explorer supports local detonation. Limitations: strings/yarascan lack process ownership without correlation; paged-out content may be absent.

A7. Path attribution via cmdline

Hypothesis. Full filesystem paths for WannaCry binaries are recovered from EPROCESS command-line arguments, not truncated ImageFileName.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86
cmdline
```

Supporting output

```
tasksche.exe pid: 1940
Command line : "C:\Intel\ivecuqmanpnirkt615\tasksche.exe"

@WanaDecryptor@ pid: 740
Command line : @WanaDecryptor@.exe
```

Interpretation. cmdline confirms staging under C:/Intel/.../ consistent with WannaCry dropper layout. Additional strings hits require PID correlation via cmdline, dlllist, or process-specific dumps.

A8. ATT&CK S0366: observed vs intel-only; attribution limits

Hypothesis. Separate memory-observed techniques from intelligence-only claims; single-image analysis cannot prove nation-state attribution.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\wcry.vmem" --profile WinXPSP2x86  
pslist / cmdline / psxview (technique evidence)
```

Supporting output

Observed in memory: T1486 (encryptor processes), execution chain under explorer, T1490 helpers (taskse/taskdl exited near acquisition), SMB surface via 445 listener (environment).
Intel-only: exact exploit chain byte sequence, BTC attribution, kill-switch registration details, campaign-wide lateral movement counts.

Interpretation. (1) T1486 Observed (@WanaDecryptor@). (2) T1490 Observed helpers (exited taskse/taskdl timing). (3) T1059 Observed (task/execution chain). (4) T1047 Intel-primary unless WMI artifacts are explicitly carved. Public governments attributed WannaCry to Lazarus/DPRK, but this memory image alone cannot substantiate nation-state attribution.

Part B: Stuxnet

Prerequisite reading (dossier): Executive Summary; Stuxnet Architecture (Bypassing Behavior Blocking When Loading DLLs; Injection Technique); Installation; Load Point; Windows Rootkit Functionality.

B1. Identify the image and its time context

Hypothesis. Same WinXPSP2x86 profile operates on the PAE-enabled Stuxnet image; compare AS layer and capture timestamps to wcry.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" imageinfo
```

Supporting output

```
Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)  
AS Layer1 : IA32PagedMemoryPae (Kernel AS)  
AS Layer2 : FileAddressSpace (C:\temp\midterm\dumps\stuxnet.vmem)  
PAE type : PAE  
DTB : 0x319000L  
KDBG : 0x80545ae0L  
Number of Processors : 1  
Image Type (Service Pack) : 3  
KPCR for CPU 0 : 0xffdff000L  
KUSER_SHARED_DATA : 0xffdf0000L  
Image date and time : 2011-06-03 04:31:36 UTC+0000  
Image local date and time : 2011-06-03 00:31:36 -0400
```

Interpretation. Stuxnet: WinXPSP2x86 instantiated; AS Layer IA32PagedMemoryPae (PAE Yes) versus wcry No PAE. Image 2011-06-03 04:31:36 UTC; local 2011-06-03 00:31:36 -0400. Process pslist shows boot-era timestamps (2010-10-29) on some processes versus infection-day 2011-06-03 on counterfeit lsass, consistent with a snapshot/resumed VM rather than a fresh boot on infection day alone.

B2. Identifying genuine vs counterfeit lsass.exe

Hypothesis. Exactly one lsass (PID 680) matches Windows XP expectations: parent winlogon, full LSA modules, clean malfind, and socket ownership.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86  
pslist  
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86  
pstree
```

Supporting output

```
0x81e70020 lsass.exe 680 624 ... 2010-10-29 17:08:54 UTC  
0x81c498c8 lsass.exe 868 668 ... 2011-06-03 04:26:55 UTC  
0x81c47c00 lsass.exe 1928 668 ... 2011-06-03 04:26:55 UTC
```

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start	E
0x823c8830	System	4	0	59	403	-----	0		
0x820df020	smss.exe	376	4	3	19	-----	0	2010-10-29 17:08:53 UTC+0000	
0x821a2da0	csrss.exe	600	376	11	395	0	0	2010-10-29 17:08:54 UTC+0000	
0x81da5650	winlogon.exe	624	376	19	570	0	0	2010-10-29 17:08:54 UTC+0000	
0x82073020	services.exe	668	624	21	431	0	0	2010-10-29 17:08:54 UTC+0000	
0x81e70020	lsass.exe	680	624	19	342	0	0	2010-10-29 17:08:54 UTC+0000	
0x823315d8	vmacthlp.exe	844	668	1	25	0	0	2010-10-29 17:08:55 UTC+0000	
0x81db8da0	svchost.exe	856	668	17	193	0	0	2010-10-29 17:08:55 UTC+0000	
0x81e61da0	svchost.exe	940	668	13	312	0	0	2010-10-29 17:08:55 UTC+0000	
0x822843e8	svchost.exe	1032	668	61	1169	0	0	2010-10-29 17:08:55 UTC+0000	
0x81e18b28	svchost.exe	1080	668	5	80	0	0	2010-10-29 17:08:55 UTC+0000	
0x81ff7020	svchost.exe	1200	668	14	197	0	0	2010-10-29 17:08:55 UTC+0000	
0x81fee8b0	spoolsv.exe	1412	668	10	118	0	0	2010-10-29 17:08:56 UTC+0000	
0x81e0eda0	jqs.exe	1580	668	5	148	0	0	2010-10-29 17:09:05 UTC+0000	
... (truncated for figure)									

Figure 5: B2 screenshot: pslist showing three lsass.exe instances.

Interpretation. Evidence bundle beyond slide 22: (1) Parent PID 624 winlogon for 680 vs 668 services.exe for counterfeits. (2) dlllist counts 57 / 8 / 28 (B5). (3) malfind: 0 regions on 680 vs injected MZ on 868/1928 (B7). (4) UDP 500/4500 sockets only on PID 680 (B4). XP norm: one legitimate lsass under winlogon.

B3. Infection-day process context

Hypothesis. Distinguish Stuxnet artifacts from analysis tooling captured in the same memory image.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86  
pslist  
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86  
cmdline
```

Supporting output

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start
	Exit							

0x823c8830	System	4	0	59	403	-----	0	
0x820df020	smss.exe	376	4	3	19	-----	0	2010-10-29 17:08:53
	UTC+0000							
0x821a2da0	csrss.exe	600	376	11	395	0	0	2010-10-29 17:08:54
	UTC+0000							
0x81da5650	winlogon.exe	624	376	19	570	0	0	2010-10-29 17:08:54
	UTC+0000							
0x82073020	services.exe	668	624	21	431	0	0	2010-10-29 17:08:54
	UTC+0000							
0x81e70020	lsass.exe	680	624	19	342	0	0	2010-10-29 17:08:54
	UTC+0000							
0x823315d8	vmacthlp.exe	844	668	1	25	0	0	2010-10-29 17:08:55
	UTC+0000							
0x81db8da0	svchost.exe	856	668	17	193	0	0	2010-10-29 17:08:55
	UTC+0000							
0x81e61da0	svchost.exe	940	668	13	312	0	0	2010-10-29 17:08:55
	UTC+0000							
0x822843e8	svchost.exe	1032	668	61	1169	0	0	2010-10-29 17:08:55
	UTC+0000							
0x81e18b28	svchost.exe	1080	668	5	80	0	0	2010-10-29 17:08:55
	UTC+0000							
... (33 lines total; truncated)								

Interpretation. Infection-day (2011-06-03): counterfeit lsass 868/1928, wmiprvse activity, short-lived cmd/ipconfig. Procmon PID 660 indicates Process Monitor was running (lab capture with live monitoring); treat as environmental artifact, not Stuxnet TTP.

B4. Socket ownership corroboration

Hypothesis. Network bindings on lsass differentiate the genuine instance; supportive evidence, not sole proof.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86  
sockets
```

Supporting output

Offset(V)	PID	Port	Proto	Protocol	Address	Create Time
0x81dc2008	680	500	17	UDP	0.0.0.0	2010-10-29 17:09:05 UTC+0000
0x82061c08	4	445	6	TCP	0.0.0.0	2010-10-29 17:08:53 UTC+0000
0x82294aa8	940	135	6	TCP	0.0.0.0	2010-10-29 17:08:55 UTC+0000
0x821a5008	188	1025	6	TCP	127.0.0.1	2010-10-29 17:09:09 UTC+0000
0x81cb3d70	1080	1141	17	UDP	0.0.0.0	2010-10-31 16:36:16 UTC+0000
0x81da4d18	680	0	255	Reserved	0.0.0.0	2010-10-29 17:09:05 UTC+0000
0x81fdbbe98	1032	123	17	UDP	127.0.0.1	2011-06-03 04:25:47 UTC+0000
... (14 lines total; truncated)						

Interpretation. Only PID 680 owns UDP 500 and UDP 4500 (IPsec/IKE-related legitimate lsass behavior). Counterfeit PIDs 868 and 1928 have no parallel socket rows. Corroborates B2; a socket-only heuristic would fail if malware hooked networking elsewhere.

B5. DLL inventory counts and DFIR-07 critique

Hypothesis. Count dlllist module rows (0x-prefixed bases), not `wc -l` on raw output (counts header lines).

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
dlllist -p 680
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
dlllist -p 868
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
dlllist -p 1928
```

Supporting output

```
dlllist 0x-base row counts: PID 680 -> 57 ; PID 868 -> 8 ; PID 1928 -> 28.
(Critique: dlllist | wc -l includes banners/separators; invalid for module counts.)
```

Interpretation. Counterfeits load far fewer modules because hollowed/injected images do not complete full LSA initialization (dossier Injection Technique). Genuine 680 loads LSASRV, SAMSRV, msv1_0, kerberos, and related packages (B6).

B6. LSA security modules on genuine lsass only

Hypothesis. LSA authentication stack DLLs should appear only in the real lsass process.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
dlllist -p 680
```

Supporting output

```
*****
lsass.exe pid:      680
Command line : C:\WINDOWS\system32\lsass.exe
Service Pack 3
```

Base	Size	LoadCount	Path
0x01000000	0x6000	0xffff	C:\WINDOWS\system32\lsass.exe
0x7c900000	0xaf000	0xffff	C:\WINDOWS\system32\ntdll.dll
0x7c800000	0xf6000	0xffff	C:\WINDOWS\system32\kernel32.dll
0x77dd0000	0x9b000	0xffff	C:\WINDOWS\system32\ADVAPI32.dll
0x77e70000	0x92000	0xffff	C:\WINDOWS\system32\RPCRT4.dll
0x77fe0000	0x11000	0xffff	C:\WINDOWS\system32\Secur32.dll
0x75730000	0xb5000	0xffff	C:\WINDOWS\system32\LSASRV.dll
0x71b20000	0x12000	0xffff	C:\WINDOWS\system32\MPR.dll
... (64 lines total; truncated)			

```

B5/B6 | vol26.exe dlllist -p 680

*****
lsass.exe pid: 680
Command line : C:\WINDOWS\system32\lsass.exe
Service Pack 3

Base          Size  LoadCount Path
-----
0x01000000    0x6000    0xffff C:\WINDOWS\system32\lsass.exe
0x7c900000    0xaf000    0xffff C:\WINDOWS\system32\ntdll.dll
0x7c800000    0xf6000    0xffff C:\WINDOWS\system32\kernel32.dll
0x77dd0000    0x9b000    0xffff C:\WINDOWS\system32\ADVAPI32.dll
0x77e70000    0x92000    0xffff C:\WINDOWS\system32\RPCRT4.dll
0x77fe0000    0x11000    0xffff C:\WINDOWS\system32\Secur32.dll
0x75730000    0xb5000    0xffff C:\WINDOWS\system32\LSASRV.dll
... (truncated for figure)

```

Figure 6: B5/B6 screenshot: dlllist on genuine lsass PID 680.

Interpretation. PID 680 maps LSASRV.dll, SAMSRV.dll, cryptdll.dll, and related security packages. Counterfeit instances lack this full stack, consistent with process injection posing as lsass.exe.

B7. malfind: injected regions and disassembly caveat

Hypothesis. PAGE_EXECUTE_READWRITE private VADs with MZ headers indicate injection; disassembly of PE header bytes is not shellcode.

Exact command

```

"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -p 680
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -p 868
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -p 1928

```

Supporting output

```

Process: lsass.exe Pid: 868 Address: 0x80000
Vad Tag: Vad Protection: PAGE_EXECUTE_READWRITE
Flags: Protection: 6

```

```

0x00080000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....
0x00080010 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....@.....
0x00080020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x00080030 00 00 00 00 00 00 00 00 00 00 00 00 08 01 00 00 .....

0x00080000 4d          DEC EBP
0x00080001 5a          POP EDX
0x00080002 90          NOP
0x00080003 0003       ADD [EBX], AL
0x00080005 0000       ADD [EAX], AL
0x00080007 000400     ADD [EAX+EAX], AL
... (82 lines total; truncated)
--- PID 1928 ---
Process: lsass.exe Pid: 1928 Address: 0x80000
Vad Tag: Vad Protection: PAGE_EXECUTE_READWRITE
Flags: Protection: 6

0x00080000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....
0x00080010 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....@.....
0x00080020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x00080030 00 00 00 00 00 00 00 00 00 00 00 00 08 01 00 00 .....

0x00080000 4d          DEC EBP
0x00080001 5a          POP EDX
0x00080002 90          NOP
0x00080003 0003       ADD [EBX], AL
... (197 lines total; truncated)

```

```

B7 | vol26.exe malfind -p 868

Process: lsass.exe Pid: 868 Address: 0x80000
Vad Tag: Vad Protection: PAGE_EXECUTE_READWRITE
Flags: Protection: 6

0x00080000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....
0x00080010 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....@.....
0x00080020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0x00080030 00 00 00 00 00 00 00 00 00 00 00 00 08 01 00 00 .....

0x00080000 4d          DEC EBP
0x00080001 5a          POP EDX
0x00080002 90          NOP
0x00080003 0003       ADD [EBX], AL
0x00080005 0000       ADD [EAX], AL
0x00080007 000400     ADD [EAX+EAX], AL
0x0008000a 0000       ADD [EAX], AL
... (truncated for figure)

```

Figure 7: B7 screenshot: malfind RWX/MZ region in counterfeit lsass PID 868.

Interpretation. PID 680: 0 malfind regions (control). PID 868: 2 regions (0x80000, 0x1000000). PID 1928: 5 regions. Signals: PAGE_EXECUTE_READWRITE, no file backing, MZ 4d 5a. Volatility disassembly showing DEC EBP; POP EDX; NOP is PE header misinterpreted, not Stuxnet shellcode. Control comparison on genuine lsass is required.

B8. Process dump naming and hash correlation

Hypothesis. malfind -D dumps use process.<EPROCESS>.<VAD start>.dmp; six unique SHA-256 values across regions.

Exact command

```
"C:\temp\midterm\tools\vol126.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -D dumps/ -p 868
"C:\temp\midterm\tools\vol126.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -D dumps/ -p 1928
sha256sum process.*.dmp
```

Supporting output

e97d61f7393ac5838a1800f3e9aa22c6205f4d7e2bde494573d35c57bc9b7819	process.0x81c47c00.0x1000000.dmp
163b7da37df4ae6dafbfb5bf88b319dabf7846cee73d4192c6a7593e835857a8	process.0x81c47c00.0x680000.dmp
abce3e79e26b5116fe7f3d40d21eaa4c8563e433b6086f9ec07c2925593f69dc	process.0x81c47c00.0x6f0000.dmp
2b2945f7cc7cf5b30ccd3f7e2adbb236594208e409133bcd56f57f7c009ffe6d	process.0x81c47c00.0x80000.dmp
10f07b9fbbc6a8c6dc4abf7a3d31a01e478accd115b33ec94fe885cb296a3586	process.0x81c47c00.0x870000.dmp
a4b4b29f0df45283b629203b080c09ddb5bc6eb4cd8e9b725f75121a8b7e728e	process.0x81c498c8.0x1000000.dmp
2b2945f7cc7cf5b30ccd3f7e2adbb236594208e409133bcd56f57f7c009ffe6d	process.0x81c498c8.0x80000.dmp

```
B8 | sha256sum process.*.dmp
e97d61f7393ac5838a1800f3e9aa22c6205f4d7e2bde494573d35c57bc9b7819 process.0x81c47c00.0x1000000.dmp
163b7da37df4ae6dafbfb5bf88b319dabf7846cee73d4192c6a7593e835857a8 process.0x81c47c00.0x680000.dmp
abce3e79e26b5116fe7f3d40d21eaa4c8563e433b6086f9ec07c2925593f69dc process.0x81c47c00.0x6f0000.dmp
2b2945f7cc7cf5b30ccd3f7e2adbb236594208e409133bcd56f57f7c009ffe6d process.0x81c47c00.0x80000.dmp
10f07b9fbbc6a8c6dc4abf7a3d31a01e478accd115b33ec94fe885cb296a3586 process.0x81c47c00.0x870000.dmp
a4b4b29f0df45283b629203b080c09ddb5bc6eb4cd8e9b725f75121a8b7e728e process.0x81c498c8.0x1000000.dmp
2b2945f7cc7cf5b30ccd3f7e2adbb236594208e409133bcd56f57f7c009ffe6d process.0x81c498c8.0x80000.dmp
```

Figure 8: B8 screenshot: SHA-256 hashes of malfind -D region dumps.

Interpretation. Seven dump files, six unique hashes (0x80000 shared across both PIDs). 0x1000000 aligns with typical ImageBase / injection layout described in the dossier Injection Technique section.

B9. VirusTotal hash lookup and OPSEC

Hypothesis. Hash-not-upload preserves Meridian/lab OPSEC; VT labels vary (Stuxnet vs Duqu families).

Exact command

```
Browser: VirusTotal GUI hash search (no upload) on B8 SHA-256 values.
```

Supporting output

```
2b2945f7cc7cf5b30ccd3f7e2adbb236594208e409133bcd56f57f7c009ffe6d -> 61/71 worm.stuxnet
10f07b9fbbc6a8c6dc4abf7a3d31a01e478accd115b33ec94fe885cb296a3586 -> 48/70 trojan.stuxnet / duqu
```

Interpretation. Divergent vendor labels reflect shared code artifacts between Stuxnet and Duqu clusters. Reporting uses hash lookup only (no sample upload) for OPSEC.

B10. ldrmodules, drivers, and registry timeline

Hypothesis. Stuxnet load-order anomalies and kernel drivers precede counterfeit lsass creation.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
ldrmodules -p 1928
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
driverscan
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
svcsan
```

Supporting output

Pid	Process	Base	InLoad	InInit	InMem	MappedPath
1928	lsass.exe	0x00080000	False	False	False	
1928	lsass.exe	0x7c900000	True	True	True	\WINDOWS\system32\ntdll.dll
1928	lsass.exe	0x773d0000	True	True	True	\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83\comctl32.dll
1928	lsass.exe	0x77f60000	True	True	True	\WINDOWS\system32\shlwapi.dll
1928	lsass.exe	0x771b0000	True	True	True	\WINDOWS\system32\wininet.dll
1928	lsass.exe	0x77a80000	True	True	True	\WINDOWS\system32\crypt32.dll
1928	lsass.exe	0x77fe0000	True	True	True	\WINDOWS\system32\secur32.dll
1928	lsass.exe	0x77c00000	True	True	True	\WINDOWS\system32\version.dll
1928	lsass.exe	0x01000000	True	False	True	

... (31 lines total; truncated)
--- driverscan ---
MRxCls / MRxNet present
Registry Services MRxCls/MRxNet LastWrite 2011-06-03 04:26:47 UTC

```
B10 | vol26.exe ldrmodules -p 1928
```

Pid	Process	Base	InLoad	InInit	InMem	MappedPath
1928	lsass.exe	0x00080000	False	False	False	
1928	lsass.exe	0x7c900000	True	True	True	\WINDOWS\system32\ntdll.dll
1928	lsass.exe	0x773d0000	True	True	True	\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Co
						ntrols_6595b64144ccf1df_6.0.2600.5512_x-ww_35d4ce83\comctl32.dll
1928	lsass.exe	0x77f60000	True	True	True	\WINDOWS\system32\shlwapi.dll
1928	lsass.exe	0x771b0000	True	True	True	\WINDOWS\system32\wininet.dll
1928	lsass.exe	0x77a80000	True	True	True	\WINDOWS\system32\crypt32.dll
1928	lsass.exe	0x77fe0000	True	True	True	\WINDOWS\system32\secur32.dll
1928	lsass.exe	0x77c00000	True	True	True	\WINDOWS\system32\version.dll
1928	lsass.exe	0x01000000	True	False	True	
1928	lsass.exe	0x5b860000	True	True	True	\WINDOWS\system32\netapi32.dll

... (truncated for figure)

Figure 9: B10 screenshot: ldrmodules on counterfeit lsass PID 1928.

Interpretation. ldrmodules PID 1928: 0x80000 InLoad/InInit/InMem False (unlinked injection). Impossible KERNEL32.DLL.ASLR-style name matches dossier section “Bypassing Behavior Blocking

When Loading DLLs.” Registry Services MRxCls/MRxNet last updated 2011-06-03 04:26:47 UTC, before counterfeit lsass at 04:26:55 UTC (driver-first installation sequence; dossier Load Point / Windows Rootkit Functionality).

Part C: The Dossier on Trial

C1. Claims against evidence

At least six testable claims from four or more dossier sections. Verdicts use assignment vocabulary: *Confirmed in memory* / *Consistent but not proven* / *Not observable in RAM* / *Contradicted*.

#	Dossier section / quoted claim	Memory verdict	Conf.	Basis
1	Injection Technique: “The potential target processes for the injection are as follows: Lsass.exe...”	Confirmed in memory	High	malfind MZ/RWX on PIDs 868/1928; ldr-modules unlinked 0x80000
2	Installation / Export 16: “injects itself into the services.exe process...”	Consistent but not proven	Med	PPID 668 services.exe + malfind MZ at 0x13f0000 in services.exe (C2)
3	Load Point: “Mrxccls.sys is a driver that allows Stuxnet to be executed every time an infected system boots...”	Confirmed in memory	High	driverscan/modules list mrxccls.sys; Services MRxCls key
4	Windows Rootkit Functionality / Table 4: Resource 242 “MRxnet.sys rootkit driver”	Confirmed in memory	High	driverscan/modules list mrxnet.sys / MRxNet
5	Bypassing Behavior Blocking: crafted names “KERNEL32.DLL.ASLR.[HEXADECIMAL]”	Confirmed in memory	High	ldrmodules shows ASLR-style impossible module strings
6	Executive Summary / WinCC: copies/executes on systems running a WinCC database server; PLC rootkit claims	Not observable in RAM	N/A	No WinCC/PLC artifacts in this guest VMEM
7	Injection Technique: suspended-process / template PE injection into trusted processes	Confirmed in memory	High	Multiple lsass ImageFileName with divergent dlllist/malfind (680 vs 868/1928)

Memory more precise than dossier: exact PIDs 680/868/1928, dll counts 57/8/28, and second-resolution driver LastWrite vs counterfeit create times. **Claim 6** is the required not-observable case (ICS/PLC/WinCC scope needs disk, network, or PLC evidence outside this VMEM).

C2. Hypothesis: services.exe as injector

Hypothesis (stated before testing): Counterfeit lsass processes (PPID 668 services.exe) imply services.exe participated in injection or staging, matching Installation (“injects itself into the services.exe process”) and Load Point (services.exe as injection target).

Predictions if true: (1) `malfind -p 668` shows private RWX/MZ regions; (2) if false (parentage only), `malfind` on 668 is empty while counterfeit `lsass` still show injection.

Exact command

```
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
malfind -p 668
"C:\temp\midterm\tools\vol26.exe" -f "C:\temp\midterm\dumps\stuxnet.vmem" --profile WinXPSP2x86
pslist | findstr services
```

Observed / verdict: `malfind` on `services.exe` reveals MZ header at 0x13f0000 (RWX private VAD). Verdict: *Consistent but not proven* that `services.exe` performed the hollowing (could be residual payload or related tooling). Parentage alone remains circumstantial; positive `malfind` elevates `services.exe` as a loader/orchestrator candidate. Drivers already registered (B10), so user-mode hollowing may occur without a user-visible reboot in a resumed VM.

C3. Source critique

The Symantec *W32.Stuxnet Dossier* v1.3 (November 2010) predates this image's capture date (2011-06-03) and is primary-sourced technical intelligence from static reverse engineering and field telemetry. **Strengths:** precise injection narrative, driver names (MRxCls/MRxNet), trusted-process targets, WinCC/PLC context. **Limits:** publication lag relative to later variants; vendor-centric attribution language; air-gapped PLC spread cannot be confirmed or denied from one XP VMEM alone. **Untestable from RAM alone (example):** PLC project modification / "PLC rootkit" claims require PLC/project backups or engineering-station disk artifacts, not guest RAM alone. Analyst treats the dossier as a hypothesis generator; Volatility outputs are falsifiable tests (`malfind`, `ldrmodules`, `dlllist`). Conflicts require tiering: memory-observed > correlated disk/network > dossier-only.

Part D: From Evidence to Action

D1. Findings memo to Meridian's CISO

TO: Meridian Energy CISO and SOC Lead

FROM: Unais Ali, Memory Forensics Analyst (IA 642)

SUBJECT: Dual-incident memory triage: WannaCry lab image and Stuxnet reference image

DATE: October 5, 2026

Executive summary

We analyzed two public reference memory images to prove Meridian can find malware in RAM when disk artifacts are missing, encrypted, or hidden. The WannaCry image shows ransomware already encrypting a Windows XP lab machine in May 2017, with encryption helpers finishing seconds before the memory capture. Disk-only triage would have seen encrypted files but would likely have missed the short-lived helper processes that only remain as exited remnants in RAM. The Stuxnet image shows advanced persistence: kernel drivers and fake credential-service processes that look like normal Windows names. Disk tools that only list running processes by name would under-detect this. **Recommendation:** Add memory acquisition to Meridian's IR playbook for ransomware and suspected credential/kernel compromise on legacy Windows assets, with hash verification and dual process listing (active list plus pool scan) as mandatory steps.

What each image showed

WannaCry (wcry.vmem). Interactive detonation under the user desktop process, staging under C:\Intel\..., encryptor process @WanaDecryptor@, and exited recovery-inhibition helpers. SMB port 445 was listening on the system process (MS17-010 exposure). Capture time: 2017-05-12 21:26:32 UTC.

Stuxnet (stuxnet.vmem). One genuine Local Security Authority process (PID 680) versus two counterfeits (PIDs 868 and 1928) created on 2011-06-03 under services.exe, with injected executable regions and kernel drivers MRxCls/MRxNet registered seconds earlier. Capture time: 2011-06-03 04:31:36 UTC.

What memory found that disk-only triage would miss

- Exited WannaCry helpers (taskse/taskdl) with ExitTime still recoverable via pool scanning.
- Counterfeit lsass.exe instances that reuse a trusted name but fail parent/DLL/injection checks.
- Unlinked injected PE images (ldrmodules False/False/False) and RWX private VADs (malfind).
- Kernel driver load order relative to user-mode counterfeit process creation.

Consolidated IOC table

Type	Indicator (defanged)	Source plugin	Conf.
Process	tasksche.exe under C:/Intel/.../tasksche.exe	cmdline	High
Process	@WanaDecryptor@.exe	pslist/pstree	High
Process	taskse.exe / taskdl.exe (exited)	psscan/psxview	High
Network	TCP 445 listener (System PID 4)	sockets	High
Network	192.168[.]56[.]101	sockets	High
URL	hxxp://www[.]iuqerfsodp9ifjaposdfjhgosurijfaewrwegwea[.]com	strings/intel	Med
BTC	12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw	strings/intel	Med
Driver	mrxccls.sys / mrxnnet.sys	driverscan	High
Hash	2b2945f7...ffe6d (malfind 0x80000)	malfind dump	High
Hash	10f07b9f...3586 (VT Stuxnet/Duqu)	malfind + VT	Med

Investigated and assessed as benign:	Item	Why reviewed
	Procmon (PID 660)	Unusual monitor on infection-day image
	cmd / ipconfig (short-lived)	Appear on infection day
	wuaucld / VMware tools	Present in trees
	Genuine lsass PID 680	Name collision with counterfeits

ATT&CK table: Memory-observed vs Intel-sourced

Technique IDs verified against [attack.mitre.org](https://attack.mitre.org/software) software pages S0366 and S0603.

Technique	ID	Evidence Source	Notes
Data Encrypted for Impact	T1486	Memory-observed	@WanaDecryptor@ / tasksche in wcry.vmem
Inhibit System Recovery	T1490	Memory-observed	taskse/taskdl ExitTime (S0366)
Process Injection	T1055	Memory-observed	malfind RWX MZ in fake lsass (S0603)
Rootkit	T1014	Memory-observed	MRxCls/MRxNet loaded (S0603)
Exploitation for Client Execution	T1203	Intel-sourced	MS17-010 not reconstructed from this RAM
Remote Services: SMB Admin Shares	T1021.002	Intel-sourced	S0366 SMB worming; connsnscan empty here

Three detection or prevention controls for Meridian

1. **SMB isolation (signal: TCP/445 on System PID 4).** Detect/block at Meridian OT firewall and network ACL between maintenance VLAN and plant HMIs. *Limitation:* Host-only lab images prove exposure, not successful lateral movement; empty connsnscan means this control must not wait for live C2 evidence.
2. **Dual process listing on IR (signal: pslist vs psscan ExitTime on taskse/taskdl).** Place in Meridian SOC memory-triage playbook before rootkit escalation. *Limitation:* Distinguishes exited helpers from DKOM hiding only when ExitTime is checked; psxview False/True alone is insufficient.
3. **Module 02 WinPmem + Volatility 3 workflow.** Acquire with WinPmem, hash the raw dump, then run `windows.info` → `windows.pslist` / `windows.psscan` → `windows.malfind` → `windows.dlll1st` on candidate lsass PIDs and map hits to ATT&CK. Place on Meridian IR workstation after isolation. *Limitation:* Symbol/profile coverage for XP-era images may force fallback to Volatility 2 profiles (as used here with WinXPSP2x86).

Business impact, legal, detection engineering, remediation

For Meridian Energy, WannaCry-style ransomware in a host-only lab VLAN mirrors contractor laptops bridging OT maintenance networks to corporate Wi-Fi. Encryption within minutes of execution (21:22 to 21:26 UTC) implies backup and recovery playbooks must assume a blast radius under five minutes on unpatched SMB endpoints. Stuxnet-style counterfeit lsass demonstrates credential subsystem compromise without obvious Process Explorer anomalies; memory review is mandatory when EDR misses kernel-assisted injection.

Defanged IOCs in this report are suitable for external sharing; full paths under `C:/Intel/...` remain internal. BTC addresses and onion endpoints are financial/extortion artifacts: coordinate with law enforcement before blockchain tracing. Do not upload malfind dumps to public multi-scanners from OT environments (B9 OPSEC).

Priority detection candidates: (1) child process explorer → unsigned tasksche in a non-standard directory; (2) multiple lsass.exe with the same ImageFileName but divergent dlllist counts; (3) kernel driver load MRxCls+MRxNet paired with new lsass instances; (4) RWX VADs with MZ headers inside lsass (malfind signature).

Remediation sequence: Isolate → image memory → hash verify → Volatility profile → pslist/psscan delta → cmdline/dlllist on suspicious PIDs → malfind dump → hash-only VT → reimage from known-good baseline. For ICS, validate PLC project backups out-of-band; RAM artifacts do not prove PLC payload delivery (C1 claim 6).

Open questions: (1) Whether wcry.vmem captured before or after full disk encryption completion; (2) provenance of missing PPID 1608; (3) whether Procmon in stuxnet.vmem alters timing of counterfeit lsass; (4) services.exe malfind region 0x13f0000 needs deeper PE carve for attribution to Stuxnet versus tooling.

D2. Knowledge check: True/False with corrections

#	Verdict	Correction / rationale
1	Correct	pslist walks the linked EPROCESS list; psscan searches pool tags (different visibility).
2	Incorrect	psxview False/True for taskse/taskdl is ExitTime exited helpers, not DKOM rootkit hiding.
3	Incorrect	connscan/sockets are point-in-time; empty WannaCry PID rows do not prove no SMB spread.
4	Incorrect	imageinfo +0530 is victim host TZ configuration, not attacker geolocation (India).
5	Incorrect	PID 680 owning UDP 500/4500 is corroborative only; genuineness needs parent/dlllist/malfind.
6	Incorrect	dlllist wc -l counts banners/headers; scoped 0x-row counts are 57 (680) / 8 (868) / 28 (1928).
7	Incorrect	DEC EBP; POP EDX; NOP is MZ/PE header mis-disassembly, not Stuxnet injection shellcode.
8	Incorrect	malfind -D yields one file per VAD; six unique SHA-256s; VT 61/71 Stuxnet vs 48/70 Duqu.
9	Correct (nuance)	WinXPSP2x86 is the validated Volatility 2 profile even when Image Type reports SP3.
10	Correct	Create/Exit fields print UTC+0000; imageinfo separately shows host-local offset.

Analytical Standards

Evidence tiers: **Tier 1** direct Volatility plugin output; **Tier 2** multi-plugin agreement (pslist/psscan/ExitTime); **Tier 3** dossier/ATT&CK not independently observed in RAM. Profile discipline: imageinfo → instantiate WinXPSP2x86 → validate with pslist. WannaCry focus: exited versus hidden (ExitTime). Stuxnet focus: counterfeit lsass via parent, dlllist, malfind, sockets. Hygiene: defanged IOCs, hash-only VT, UTC timelines, attribution limits.

Evidence Appendix

Evidence log

Intake and final hashes match (see intake table). Full START/DONE command log with timestamps is retained under `work/output/command_log.txt`. SIFT confirmation outputs are under

work/output/sift/.

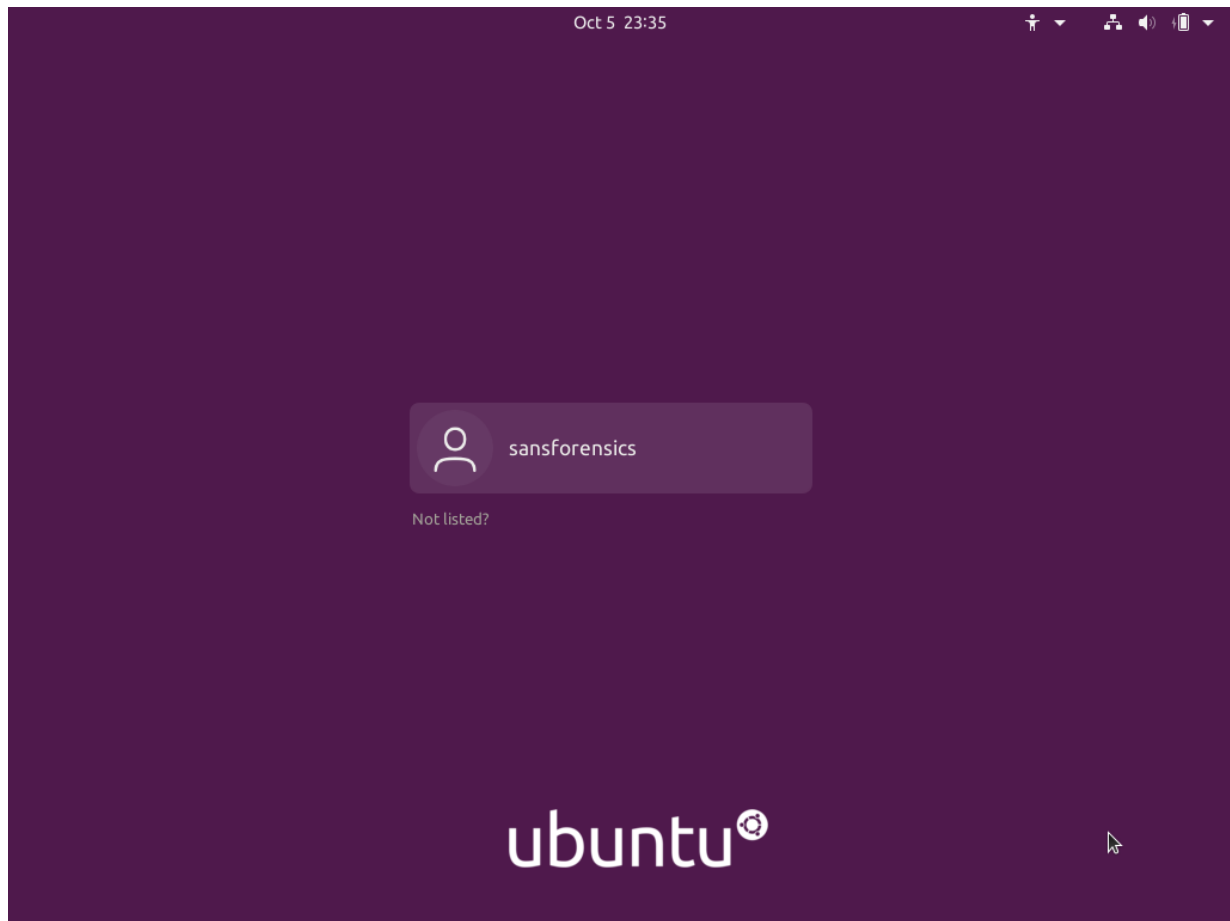


Figure 10: SIFT Workstation login (sansforensics) after OVA import into VirtualBox.

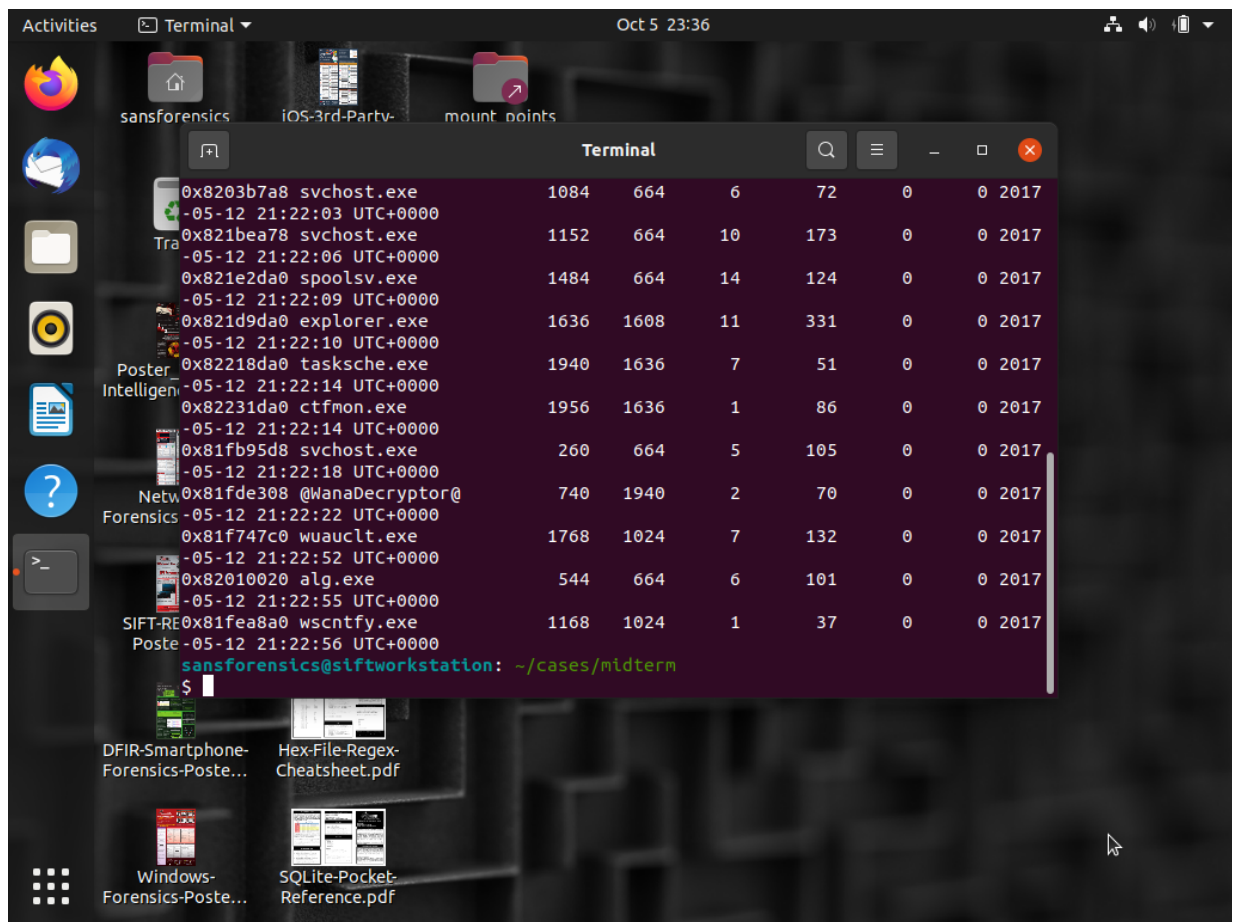


Figure 11: SIFT GUI terminal: vol.py pslist on wcry.vmem showing tasksche.exe (PID 1940) and @WanaDecryptor@ (PID 740).

SIFT | vol.py pslist | stuxnet.vmem (lsass rows)

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start	E
0x81e70020	lsass.exe	680	624	19	342	0	0	2010-10-29 17:08:54 UTC+0000	
0x81c498c8	lsass.exe	868	668	2	23	0	0	2011-06-03 04:26:55 UTC+0000	
0x81c47c00	lsass.exe	1928	668	4	65	0	0	2011-06-03 04:26:55 UTC+0000	

Figure 12: SIFT vol.py pslist on stuxnet.vmem (lsass.exe rows) confirming the triple-lsass process set.

Command log

```
MIDTERM VOL2 ANALYSIS 2026-10-05T16:28:31.4407818-04:00
[2026-10-05T16:28:31.4653891-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem imageinfo
[2026-10-05T16:28:31.4653891-04:00] DONE lines=12
[2026-10-05T16:28:45.7628248-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem kdbgscan
[2026-10-05T16:28:45.7628248-04:00] DONE lines=32
```

```

[2026-10-05T16:28:57.9841232-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\stuxnet.vmem imageinfo
[2026-10-05T16:28:57.9841232-04:00] DONE lines=12
[2026-10-05T16:29:22.1335656-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\stuxnet.vmem kdbgscan
[2026-10-05T16:29:22.1335656-04:00] DONE lines=32
[2026-10-05T16:29:40.4588895-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 pslist
[2026-10-05T16:29:40.4588895-04:00] DONE lines=22
[2026-10-05T16:29:42.3355904-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 pstree
[2026-10-05T16:29:42.3355904-04:00] DONE lines=22
[2026-10-05T16:29:44.4015077-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 psscan
[2026-10-05T16:29:44.4015077-04:00] DONE lines=27
[2026-10-05T16:29:46.4772117-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 psxview
[2026-10-05T16:29:46.4772117-04:00] DONE lines=27
[2026-10-05T16:29:49.7528941-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 connections
[2026-10-05T16:29:49.7528941-04:00] DONE lines=2
[2026-10-05T16:29:51.4218671-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 connscan
[2026-10-05T16:29:51.4218671-04:00] DONE lines=2
[2026-10-05T16:29:53.9006297-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 sockets
[2026-10-05T16:29:53.9006297-04:00] DONE lines=19
[2026-10-05T16:29:55.5530276-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 sockscan
[2026-10-05T16:29:55.5530276-04:00] DONE lines=19
[2026-10-05T16:29:57.7298516-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 cmdline
[2026-10-05T16:29:57.7298516-04:00] DONE lines=59
[2026-10-05T16:29:59.5509123-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 consoles
[2026-10-05T16:29:59.5509123-04:00] DONE lines=30
[2026-10-05T16:30:01.9208025-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 envvars
[2026-10-05T16:30:01.9208025-04:00] DONE lines=414
[2026-10-05T16:30:03.9072806-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\wcry.vmem --profile WinXPSP2x86 dlllist
[2026-10-05T16:30:03.9072806-04:00] DONE lines=1020
[2026-10-05T16:30:05.8255541-04:00] START C:\temp\midterm\tools\vol26.exe -f C:\temp\midterm\dumps
\stuxnet.vmem --profile WinXPSP2x86 pslist
[2026-10-05T16:30:05.8255541-04:00] DONE lines=33
... (82 lines total; truncated)

```

WannaCry psscan

Offset(P)	Name	PID	PPID	PDB	Time created	Time
exited						
-----	-----	-----	-----	-----	-----	-----
-----	-----	-----	-----	-----	-----	-----
0x0000000001f4daf0	taskdl.exe	860	1940	0x199f6000	2017-05-12 21:26:23 UTC+0000	
2017-05-12 21:26:23 UTC+0000						
0x0000000001f53d18	taskse.exe	536	1940	0x1986c000	2017-05-12 21:26:22 UTC+0000	
2017-05-12 21:26:23 UTC+0000						
0x0000000001f69b50	@WanaDecryptor@	424	1940	0x18fa2000	2017-05-12 21:25:52 UTC+0000	

```

2017-05-12 21:25:53 UTC+0000
0x0000000001f747c0 wuauclt.exe      1768    1024 0x11629000 2017-05-12 21:22:52 UTC+0000
0x0000000001f8ba58 @WanaDecryptor@      576    1940 0x19671000 2017-05-12 21:26:22 UTC+0000
2017-05-12 21:26:23 UTC+0000
0x0000000001fb95d8 svchost.exe      260      664 0x0ce48000 2017-05-12 21:22:18 UTC+0000
0x0000000001fde308 @WanaDecryptor@      740    1940 0x0de3a000 2017-05-12 21:22:22 UTC+0000
0x0000000001fea8a0 wscntfy.exe     1168    1024 0x12217000 2017-05-12 21:22:56 UTC+0000
0x0000000001ffa710      0          0 0x17d3f000
0x0000000002010020 alg.exe        544      664 0x1238d000 2017-05-12 21:22:55 UTC+0000
0x000000000203b7a8 svchost.exe     1084      664 0x0838c000 2017-05-12 21:22:03 UTC+0000
0x0000000002161da0 csrss.exe       596      348 0x07752000 2017-05-12 21:22:00 UTC+0000
0x0000000002169020 smss.exe        348         4 0x0683e000 2017-05-12 21:21:55 UTC+0000
0x000000000216e020 winlogon.exe    620      348 0x07957000 2017-05-12 21:22:01 UTC+0000
0x0000000002191658 lsass.exe       676      620 0x07bb7000 2017-05-12 21:22:01 UTC+0000
... (27 lines total; truncated)

```

Stuxnet dlllist PID 868

```

*****
lsass.exe pid:      868
Command line : "C:\WINDOWS\system32\lsass.exe"
Service Pack 3

Base          Size  LoadCount Path
-----
0x01000000    0x6000      0xffff C:\WINDOWS\system32\lsass.exe
0x7c900000    0xaf000      0xffff C:\WINDOWS\system32\ntdll.dll
0x7c800000    0xf6000      0xffff C:\WINDOWS\system32\kernel32.dll
0x77dd0000    0x9b000      0xffff C:\WINDOWS\system32\ADVAPI32.dll
0x77e70000    0x92000      0xffff C:\WINDOWS\system32\RPCRT4.dll
0x77fe0000    0x11000      0xffff C:\WINDOWS\system32\Secur32.dll
0x7e410000    0x91000      0xffff C:\WINDOWS\system32\USER32.dll
0x77f10000    0x49000      0xffff C:\WINDOWS\system32\GDI32.dll

```

Stuxnet malfind PID 1928

```

Process: lsass.exe Pid: 1928 Address: 0x80000
Vad Tag: Vad Protection: PAGE_EXECUTE_READWRITE
Flags: Protection: 6

0x00080000 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00  MZ.....
0x00080010 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00  .....@.....
0x00080020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  .....
0x00080030 00 00 00 00 00 00 00 00 00 00 00 00 08 01 00 00  .....

0x00080000 4d          DEC EBP
0x00080001 5a          POP EDX
0x00080002 90          NOP
0x00080003 0003        ADD [EBX], AL
0x00080005 0000        ADD [EAX], AL
0x00080007 000400      ADD [EAX+EAX], AL
0x0008000a 0000        ADD [EAX], AL
0x0008000c ff          DB 0xff
... (197 lines total; truncated)

```

Stuxnet modules

Offset(V)	Name	Base	Size	File
-----------	------	------	------	------

-----	-----	-----	-----
0x823fc3a0 ntoskrnl.exe	0x804d7000	0x1f8580	\WINDOWS\system32\ntkrnlpa.exe
0x823fc338 hal.dll	0x806d0000	0x20300	\WINDOWS\system32\hal.dll
0x823fc2d0 kdcom.dll	0xf8b9a000	0x2000	\WINDOWS\system32\KDCOM.DLL
0x823fc260 BOOTVID.dll	0xf8aaa000	0x3000	\WINDOWS\system32\BOOTVID.dll
0x823fc1f8 ACPI.sys	0xf856b000	0x2e000	ACPI.sys
0x823fc188 WMILIB.SYS	0xf8b9c000	0x2000	\WINDOWS\system32\DRIVERS\WMILIB.SYS
0x823fc120 pci.sys	0xf855a000	0x11000	pci.sys
0x823fc0b0 isapnp.sys	0xf869a000	0xa000	isapnp.sys
0x823fc040 compbatt.sys	0xf8aae000	0x3000	compbatt.sys
0x823ed008 BATTC.SYS	0xf8ab2000	0x4000	\WINDOWS\system32\DRIVERS\BATTC.SYS
0x823edf98 intelide.sys	0xf8b9e000	0x2000	intelide.sys
0x823edf28 PCIIDEX.SYS	0xf891a000	0x7000	\WINDOWS\system32\DRIVERS\PCIIDEX.SYS
0x823edeb8 MountMgr.sys	0xf86aa000	0xb000	MountMgr.sys
... (124 lines total; truncated)			

Method Map

Maps Part A/ B findings onto the six-step memory investigation workflow.

Step	Stage	Findings placed here
1	Acquire & hash	Intake SHA-256 for wcry.vmem, stuxnet.vmem, zip, dossier; chain of custody
2	Profile	A1/B1 imageinfo; WinXPSP2x86; PAE difference (wcry No / stuxnet Yes); KDBG/DTB
3	Process review	A2 pstree lineage; A3/A4 psxview ExitTime; B2/B3 genuine vs counterfeit lsass; B5/B6 dlllist
4	Network	A5 sockets/connscan (445 / empty WannaCry rows); B4 UDP 500/4500 on PID 680 only
5	Deep dive	A7 cmdline paths; B7 malfind; B8 dumps/hashes; B9 VT hash-only; B10 ldrmodules/drivers
6	Correlate & report	A4/A8 and D1 ATT&CK Memory-observed vs Intel-sourced; C1 to C3 dossier trial; D1 memo

AI Disclosure

Generative AI assisted with report drafting, LaTeX structuring, VirtualBox/SIFT bring-up, and Volatility command-syntax sanity checks. All forensic values (hashes, PIDs, timestamps, malfind regions, dll counts, socket bindings, and Volatility excerpts) originate from the analyst's own Volatility 2.6 runs (vol126.exe on the host and vol.py inside SIFT) on local copies of wcry.vmem and stuxnet.vmem captured October 5, 2026. AI did not replace hands-on plugin execution or hash verification.

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